

Task - 9

▼ Univariate Analysis on Order Level Data

▼ Order ID →

- There are 99441 order values in this dataset.
- No duplicate orders.

▼ Status →

- Most of the product are delivered according to this dataset.
- Also there are some(~600) values, data of which is labeled as unavailable, maybe these are some services provided by them.

▼ Purchase Date →

- There are details of orders for 773 days.
- There is an upwards trajectory in order placed over time.
- Approx 150 orders are places at each day on avarage.
- Most of the orders are from week days, weekend have less orders...most of the orders are from the start of the week.
- There are more orders in the start of year, than end of year.

▼ Approved Date →

- More than 69% of orders are approved in same day.

▼ Delivered Carrier Date →

- Products generally reach in 2 days on avarage.
- It takes approximately 7 days to deliver the product to the customer.

- There are some wrong entries, delivered to customer date is before delivered to carrier date.

▼ **Delivered Customer Date** →

- Carrier takes 7 days on average to deliver it to the customer.
- After 10-12 days of ordering, the product is generally delivered to the customer.

▼ **Estimated Delivery Date** →

- Company estimates to deliver items on average 23 days.

▼ **Customer ID** →

- There are 96096 customers, who have ordered something.
- There is one customer with 17 orders.

▼ **Customer City** →

- sao paulo have large number of users among other cities, 15% of overall cities.

▼ **Customer State** →

- SP have the most number of customer, approx 42% of overall customer among 27 states.

▼ **Order Item ID** →

- Most people are ordering a single item in a certain order, approx 90%.

▼ **Product ID** →

- There are 99441 unique products that have been ordered at least once.

▼ **Price** →

- 99 percentile values of price falls under 1000.
- There are only 6 orders more than 5000 and one order falls after 10000.

▼ **Seller ID** →

- There are 3084 unique sellers from whom something have ordered.
- Median order count per seller is 6.

▼ Freight Value →

- Approx 99 Percentile freight values are under 100.
- Average freight value is 17

▼ Sequential Payment →

- More than 99% of people pay in one go or use 2 sequence.

▼ Payment Types →

- 75% of people use credit card for payment.
- There are 4 types of payments accepted.
- 3 of the cancelled payments have undefined payment types, as they were not paid.

▼ Installments →

- Half of the people like to pay without installments.
- There are 2 orders with 0 payment installmens, 1 is the standard payment installment value for single payment.

▼ Seller City →

- Sao paulo have most sellers as well as most customers (~25% of all sellers).

▼ Seller State →

- SP Have the most number of sellers too as well as customers (~70% of all sellers).

▼ Product Category →

- There are 71 unqiue product category, but 3 product category holds 26% of the sold products.(bed bath table, health beuty, sports leusire)

▼ Purchase Time →

- Most of the people order from 12 PM to 12 AM.

▼ Approve Time →

- Company approves the product throughout the day, but very less at early morning.

▼ Delivered Carrier Time →

- Orders are more frequent to delivered to carrier in day time, from 12PM to 12AM.

▼ Delivered Customer Time →

- Same goes with customer delivery, happen in day time, but there are also approx 8% delivery done in midnight.

▼ Bivariate Analysis on Order Level

▼ Date Analysis

- After a sudden drop at first and sudden peak at last... the median order price is in same range throughout the time.
- Total order price is increasing over the time.



Order amount is increasing over the time, means the business is in good state from the order presepective.

- Total order count is also increasing over time.
- There is no such pattern in cancellation over time.
- There is a sudden high percentage of cancellation in the last.
- Number of customers are increasing over the years.



Graph of orders and customers are kind of same, so we can say most of the orders in each month is done by new customers

- Delivery time have reduced from the first, and also it is improving slightly at the last.
- New Cities are adding till now, lesser in numbers...but still there is growth.
- Freight value is changing at the last.
- There is no pattern in sales in particular time of year, or particular day.
- Number of new sellers are still increasing.



Median Purchase is same throughout all the year, days of week and days of months...so there is no such peak time.

▼ Categorical to Categorical

- diapers_and_hygiene and dvds_blu_ray have a higher chance of cancellation among others.
- Office furniture have the lowest chance of cancellation.
- Payment installments are not influenced by the category.
- Orders with higher order item id is not likely to cancel.
- bed bath table is the most purchased category but health_beauty is the most purchased category in early morning.
- Orders with higher order_item_id is more likely to delivered at evening.

- All the product category is ordered using the credit card most as a payment method.
- Three of the product category is highest for 4 different payment types.
- Products ordered in the early morning have the highest cancellatio rate.

▼ Numerical to Numerical

- There is no connection between freight value and price.

▼ Categorical to Numerical

- There are some high valued orders which are cancelled.
- Cancelled orders have the lowest freight values.
- Orders with more than 10 items have variable price, can not be determined.
- Orders with more than 9 order items, dont have any specific freight value pattern.
- There is no difference in mean among orders where sequential payment is done and not done.
- Credit card have relatively higher payment amount.
- Higher Delivery time have lesser freight value compared to lower delivery time.
- Orders with highest median payment value is delivered at midnight.